SECP 2613

SYSTEM ANALYSIS AND DESIGN

ASSIGNMENT 2:

PROJECT MANAGEMENT

LECTURER:

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SUBMITTED BY:

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SEMESTER II (2023/2024)

Q1

a. How would you assess the feasibility of the systems project Ahmad is proposing? (2 Marks)

> To assess the feasibility of the systems project Ahmad is proposing, I shall look at the current system technology, the problem of it and what will be improved of the proposed system. I will also look for the potential benefits and disruptions when implementing the new system. It is also important to consider what managers, system people and potential user's say before implementing the new system.

b. Based on what Abu has said about the managers, users, and systems people, what seems to be the operational feasibility of the proposed project? (2 Marks)

> Based on Abu's feedback, the operational feasibility seems low. If the proposed system is performing the way it should and potential users don't see any problems with it, the proposed system might be very disruptive for the company. As a result, while the technology may be there to implement a new system, the question remains whether it is aligned with their objectives and needs. This means the new system may not be used much.

c. What about the economic feasibility? (2 Marks)

> For economic feasibility, Abu mentioned concerns about the cost of the new system to make sure the proposed project will give the business a good return on its investment. So, it is important to do a Cost Benefit Analysis beforehand.

d. What about the technological feasibility? (2 Marks)

> Technological feasibility needs to be done to make sure people in the system have the right tool to implement the proposed system. Although based on Abu's feedback for the technology to implement the proposed system is available, he also mentioned whether changes need to be made to the current system.

e. Based on what Ahmad and Abu have discussed, would you recommend that a full-blown systems study be done? Discuss your answer in a paragraph. (2 Marks)

> Based on what they discussed, I would say I do not recommend a full-blown system. Regarding Abu's feedback, even though the technology for the system is available, there are still doubts in implementing the system because the current users are satisfied with the current system. Doing a full analysis about the cost and benefit for the proposed system is recommended to determine whether they want to pursue a full-blown system.

Q2 Work Base Schedule (WBS)

Activity No	Task	Estimated Duration (Days)	Predecessor
A	Define project scope and objectives	3	None
B	Identify stakeholders	2	None
C	Create project charter	4	A
D	Research existing student registration system	5	C
E	Review best practices in student registration systems	4	D
F	Gather requirements from students and faculty	6	E
G	Analyze requirements and prioritize features	5	F
H	Determine hardware and software requirements	4	G
I	Evaluate build and buy options	4	H
J	Select development approach	3	I
K	Estimate budget for development and implementation processes	5	G, J
L	Develop proposal document	7	K
M	Prepare and present proposal for stakeholders	4	L
N	Gather feedback from stakeholders	2	M
O	Evaluate feedback	4	N
P	Finalize proposal and submit for approval	4	O

Q3

Key Differences in Managing Projects Using Traditional Methodology and Agile Methodology

I. Introduction

The traditional approach a system analyst would use in projects is known as System Development Life Cycle (SDLC). Traditional SDLC consists of seven phases which are identifying problems, determining human information requirements, analysing system needs, designing the system, developing and documenting software, testing and maintaining system and finally, implementing and evaluating the system. On the other hand, agile methodology is an alternative to the traditional SDLC. Agile Modelling Development Process consists of five phases which are exploration, planning, iterations to the first release, productionizing and maintenance. The purpose of this report is to discuss the key differences faced by a manager using traditional methodology and agile methodology when project team members are scattered across different time zones and in different physical locations.

II. Key Differences :

a. Duration of Project

In traditional approaches, the project team members are mostly at the same physical location and the time zones are same. Thus, the project members tend to work together and complete the project in much shorter period compared to agile methodology especially when the project team members are scattered. Hence, in agile methodology, the expected due date of the project is much longer than the traditional ones as each member has to settle their parts and test it to check its usability.

b. Bug Identification

In traditional approaches, the design is created first and then tested, where the possibility of finding bugs is much higher whereas in agile methodology, the product is tested from time to time. Therefore, the system analysts that use agile methodology do not have to resolve a lot of bugs at once.

c. Communication between Team Members

In traditional approaches, team members often meet each in person where meetings are scheduled together in a certain place whereas in agile methodology, online meetings using platforms such as Microsoft Teams, Webex, Zoom, etc. occurs. Hence, arranging meetings to discuss the progress of the project is much easier using agile methodology compared to traditional approach as the hustle of finding a location reduces.

d. *Involvement of Customers*

In traditional approaches, target customers or clients are interviewed earlier on so that the system analyst knows the needs and wants of the client whereas in agile methodology, the customers get to be part of the process and not just interviewed once. Therefore, the customers can give their feedback on the projects' process from time to time and the manager can make sure that their meeting the clients' needs.

III. Conclusion

Having team members dispersed in different locations and time zones is challenging for a manager. However, using the correct methodology can ease the work process and in this case, using agile methodology can result in a better work progress compared to traditional approach.

Q4 (a)

Total Development cost = RM(50000+15000+30000+80000+40000) = RM215000
Production cost (Year 1) = RM(15000+75000+10000) = RM100000
Production cost (Year 2) = RM(15000+(75000(1.1))+10000) = RM107500
Production cost (Year 3) = RM(15000+(82500(1.1)+10000) = RM115750

Benefits for 3 years;
Year 1:

- Improve customer service =RM150000
- Increase productivity = RM150000
- Total = RM300000

Year 2:

- Improve customer service =RM150000(1.2) = RM 180000
- Increase productivity = RM150000(1.25) = RM187500
- Total = RM367500

Year 3:

- Improve customer service =RM180000(1.2) = RM216000
- Increase productivity = RM187500(1.25) = RM234375
- Total = RM450375

Calculation of present value(PV):

PV = FV / (1 + r)^n

Present Value for costs:
PV = 215000+(100000/(1+0.35)^1)+(107500/(1+0.35)^2)+(115750/(1+0.35)^3)
PV = 215000+74074.07+58984.91+47045.67
PV = 395104.65

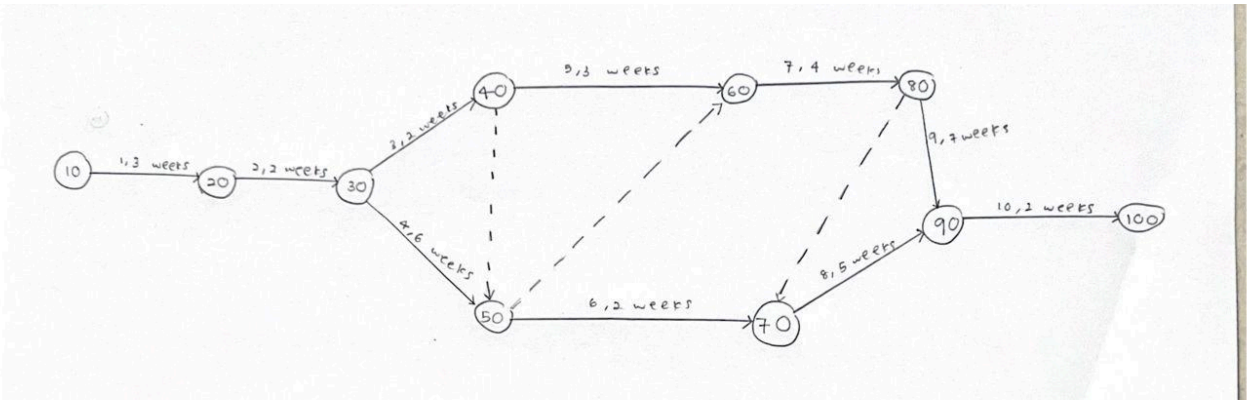
Present Value for benefits;
PV = 300000/(1+0.35)^1+367500/(1+0.35)^2+450375/(1+0.35)^3
PV = 222222.22+201646.09+183051.36
PV = 606919.67

	Year 0	Year 1	Year 2	Year 3
Development Cost				
Hardware	50000			
Software	15000			
Training	30000			
Consulting	80000			
Data Conversion	40000			
Total	215000			
Production Cost				
Supplies		15000	15000	15000
IS Salaries		75000	82500	90750
Upgrades		10000	10000	10000
Total		100000	107500	115750
Benefits				
Improve customer service		150000	180000	216000
Increase productivity		150000	187500	234375
Total		300000	367500	450375
Present Value				
Development Cost	215000			
Production Cost		74074	58985	47046
Benefits		222222	201646	183051
Accumulated Costs	215000	289074	348059	395105
Accumulated Benefits		222222	423868	606919
Gain or Loss		-66852	75809	211814
Profitability Index	0.99			

(b) Profitability index is 0.99, suggesting/ that the project is slightly below par since the index is less than one. Thus, it is not a good investment

Q5

a)



b)

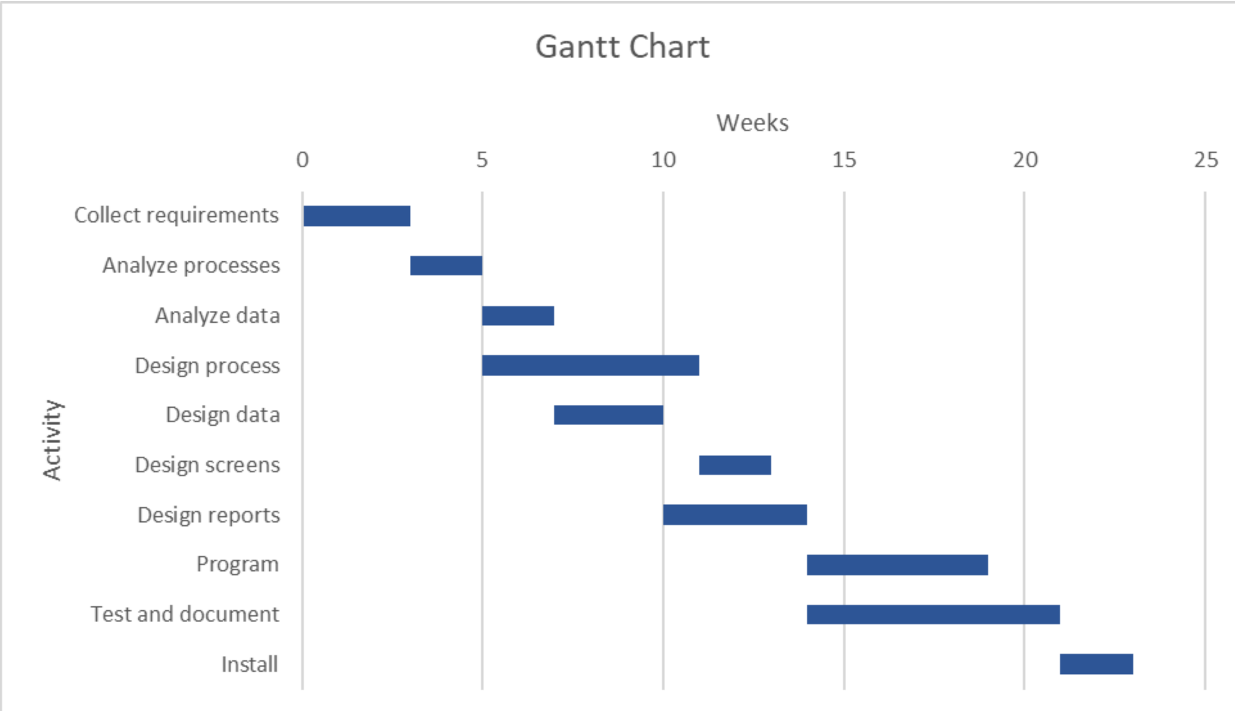
- 1-2-3-5-7-8-10 (Path 1)
- 1-2-3-5-7-9-10 (Path 2)
- 1-2-3-6-8-10 (Path 3)
- 1-2-4-7-8-10 (Path 4)
- 1-2-4-7-9-10 (Path 5)
- 1-2-4-6-8-10 (Path 6)

c)

- Path 1 : 3+2+2+3+4+5+2 = 21 weeks
- Path 2 : 3+2+2+3+4+7+2 = 23 weeks
- Path 3 : 3+3+3+3+5+2 = 16 weeks
- Path 4 : 3+2+6+4+5+2 = 22 weeks
- Path 5 : 3+2+6+4+7+2 = 24 weeks
- Path 6 : 3+2+6+2+5+2 = 20 weeks

The critical path is path 5 with duration of 24 weeks. Path 5 is 1-2-4-7-9-10. The activities in this path are collecting requirements, analyzing processes, designing processes, designing reports, test and documentation and install.

d)


Gantt Chart
Weeks
0
5
10
15
20
25
Activity
Collect requirements
Analyze processes
Analyze data
Design process
Design data
Design screens
Design reports
Program
Test and document
Install